2026/27 :: Data Visualization :: Project Description

Project start: 26th Oct, in class.
From then on, a ZIP of the project folder and a progress check are submitted on Moodle every Friday @ 23:30 (from 30th Oct to 11th Dec).

Project proposal (PPA) submission deadline: 6th Nov @ 23:30 (inside the ZIP of checkpoint 1)

Final article (PD) submission deadline: 11th Dec @ 23:30 (inside the ZIP of checkpoint 4)

Final project presentation: 14th Dec, from 14:00 to 18:00 (12 groups, 12+5 minutes each, with a 10-minute break after the sixth group).

Important:

- Be aware of the University of Porto Policy on Plagiarism.
- Properly cite any external source that you used.
- You are **not allowed to** use or reuse any piece of code **created by other groups**.
- The work must be developed in R, with the tools taught in the course (ggplot2, dplyr, plotly and Shiny). You can use code that is freely available (open source) on the Internet, provided you understand it and refer to it.
- Generative AI tools may be used **only within the limits** set in the section "Use of AI tools" below, and their use must be declared every week.

Main goals

This year the project uses the NYPD Complaint Data: every offence reported to the New York City Police Department (NYPD) between January 2020 and June 2026, 3 386 974 complaints in total. For each complaint you have the date and time of occurrence, the borough, the precinct and the coordinates, the type of premises (street, apartment building, subway, park, store, bar and about sixty others), the position relative to the premises (inside, in front of, opposite), the offence class and its level (felony, misdemeanor or violation), and some attributes of suspects and victims. The data file (`nypd_2020_2026.rds`) and its data dictionary are available on Moodle. All groups work on exactly the same data.

The main goal of the project is to **help the Mayor of New York City and the NYPD to reduce the number of incidents and make the city safer**. For example, counting offences by borough is only the starting point. The questions of interest are conditional: where do offences concentrate, at what hours and on which days, in what kinds of places, for which types of offence, and how have these patterns changed since 2020? The visualizations you build should turn such patterns into evidence for concrete decisions, such as where and when to reinforce patrols, which kinds of premises need prevention measures, or which types of offence are growing and where. Your readers are the Mayor and the police chief: they have little time, they are not data scientists, and each figure should show them a pattern and the decision it supports.

The project is developed in stages, following the tools taught in the course: exploratory analysis with ggplot2, from which the research questions emerge, hypotheses tested with visualizations, interactive figures with plotly, and a Shiny dashboard that brings the results together. Each stage builds on the previous one, and the weekly submissions (see below) record that growth.

Project assessment elements:

You get points based on the quality of the report/dashboard, the difficulty of the problem you are addressing and the novelty of your solution(s). The three components below add up to 60 of the 100 points of the course. The proposal and the final article are submitted in PDF, inside the weekly ZIP of the corresponding checkpoint, on Moodle (see "Weekly submissions and checkpoints"). The ZIP of checkpoint 4 should also contain the slides or screenshots that will be used for the presentation, as well as all the code used.

1. **[10 points] Project Preliminary Analysis (PPA), checkpoint 1, 6th Nov**
   a. The proposal should be an A4 document (max 4 pages), you can use 2 columns if you want, with font size 10 (main text), and figures, describing:
      1) The dataset, to show that you have a deep knowledge of it (including its gaps and oddities).
      2) The (undergone and expected) data transformations on the dataset.
      3) The research questions you chose, the exploratory visualizations (ggplot2) that led to them, and the first hypotheses you intend to test. Say what the Mayor or the NYPD could do with the answers.

2. **[25 points] Project Development Report (PD), checkpoint 4, 11th Dec**
   a. The final article should be written in the standard IEEE conference format (max 6 pages).
   b. The article must include the title, authors, an abstract, introduction and a conclusion.
   c. Clearly explain your project (what problems are you solving), discuss its importance, your solution(s) to tackle the problem and your novelties, in both the visualizations themselves and in the data processing methods to support the creation of the visualizations.
   d. The article should follow the path of the project: the questions of the proposal, the hypotheses you tested (including the ones you discarded, and why), the interactive figures and the dashboard. Every conclusion should be supported by a figure in the article or by a view of the dashboard.
   e. Clearly cite all tools, libraries, and external resources you used in your project. You need to explain what you added, or changed, to the existing code and resources.
   f. **Be advised:** it is important that your group add something (e.g., a novel question and answer in a visual form) that differentiates itself from any other group in the class.

3. **[25 points] Final Presentation (FP), 14th Dec, 14:00 to 18:00**
   a. Prepare a presentation of 12 minutes at most to demonstrate your visualizations, with a live demo of the Shiny dashboard, and the knowledge you get from them. The dashboard should be running on your laptop before your turn, so that no time is lost in setting up.
   b. The 5 minutes after the presentation go to discussing your choices with the group: why this figure and not another, why this interaction, why a hypothesis was dropped. Every member of the group should be able to take part in that discussion.
   c. With 12 groups in four hours (12 minutes of presentation, 5 of discussion and 2 to change over, plus a 10-minute break after the sixth group), the schedule has no slack: the time limit is strict, and a presentation is stopped when it reaches 12 minutes.
   d. You also get points based on the quality of the presentation (both orally and visually).

Weekly submissions and checkpoints

The project runs in class from 26th October to 14th December. Every week, until Friday at 23:30, each group submits two things on Moodle: a ZIP of the whole project folder as it stands (cumulative, with the work of all previous weeks, and without the data file), and a progress check with tick boxes, which takes one minute to fill in.

The ZIP may be replaced as many times as you want during the week, and the last one uploaded before the deadline is the one that counts.

Submissions between 23:30 and 23:59 are accepted but marked as late. After 23:59 nothing is accepted.

Four of the seven weekly submissions are checkpoints, with required content and feedback from the teacher:

Checkpoint	Deadline	Required content in the ZIP
C1. Questions and exploration	6th Nov	The research questions chosen, the exploration in ggplot2 (scripts and figures) and the PPA proposal (PDF).
C2. Hypotheses	20th Nov	The hypotheses derived from the questions, each with the figure that supports or refutes it, and the hypotheses you discarded, with the reason.
C3. Interaction	27th Nov	The figures made interactive with plotly and, for each one, a sentence on what the interaction adds.
C4. Dashboard and article	11th Dec	The Shiny app, the PD article (PDF), the slides or screenshots of the presentation and all the code.
Final presentation	14th Dec	12 minutes of demo and 5 of discussion per group, from 14:00 to 18:00.

How many figures, hypotheses and interactive views the group produces is the group's decision. The weekly record is used to confirm that the work grew week by week, at the pace of the tools taught in class. A week without submission, or with a ZIP identical to the previous one, is recorded as a week without progress. The complete rules (folder structure, file names, the items of the progress check) are in the guide "Project: weekly submissions and checkpoints", available on Moodle.

Use of AI tools

Generative AI tools are allowed in this project for three purposes only:

1. **Polishing text that you wrote**: grammar, clarity and concision of the proposal, the article and the slides. The ideas, the structure and the conclusions remain yours.
2. **Polishing figures that you built**: improving the code of an existing figure (labels, colours, theme, layout) or fixing an error in it.
3. **Data manipulation**: writing or fixing code that cleans, filters, joins, aggregates or reshapes the data (dplyr, tidyr), provided you understand the code and check its result.

They **may not** be used to:

- **Create figures or visualizations from scratch.** Deciding what to plot and how, and writing the code of a figure from scratch, is the core of this course and must be done by the group.
- **Explore the dataset**, formulate the research questions or the hypotheses, interpret the results, or write the analysis and the conclusions.
- **Invent, fabricate or alter data, numbers or results.** Every number and every figure in your documents must come from your own code, run on the project data.

Every week, in the progress check, the group declares what the tools were used for. Declared use within these limits is not penalized.